

Supplementary material for “Shrinkage invalidates the Hosmer–Lemeshow test: goodness of fit for penalized logistic regression, with an application to glaucoma diagnosis”

Ebrahim Khaled Ebrahim

Numbering of propositions matches the main text. Section S1 gives the regularity conditions and the proofs. Section S2 catalogues the neighbouring constructions that fail, with the size of each failure, because the question “why not the obvious thing?” has four different answers. Section S3 specifies the simulation designs. Section S4 gives the full grids and the sensitivity analyses. Section S6 lists the scripts and seeds.

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S1 Regularity conditions and proofs

S1.1 Conditions

Throughout, $y_i \in \{0, 1\}$ are independent given x_i , and $\pi_i(\beta) = \{1 + \exp(-x_i^\top \beta)\}^{-1}$. Write $W = \text{diag}\{\hat{\pi}_i(1 - \hat{\pi}_i)\}$, $F = X^\top W X$, $K = \lambda D$ with $D = \text{diag}(0, 1, \dots, 1)$, and $M = F + K$. We assume:

(C1) the true parameter β_0 lies in the interior of a compact parameter set, and $\pi_i(\beta_0) \in [\varepsilon, 1 - \varepsilon]$ for some $\varepsilon > 0$ uniformly in i ;

(C2) $n^{-1}F \rightarrow \Sigma$, positive definite, and λ is non-random with $\lambda = o(n)$, so that $n^{-1}M \rightarrow \Sigma$ and the shrinkage displacement $M^{-1}K\beta = o(1)$;

(C3) the grouping matrix C has G fixed, each group receiving a fixed positive fraction of the sample, and $V = CWC^\top$ is positive definite;

(C3') the group boundaries converge in probability to the population quantiles of the linear predictor at rate $O_p(n^{-1/2})$, with limiting cell probabilities bounded away from zero. This is an assumption, not something we discharge: in practice the groups are equal-frequency cells of the *fitted* index, so C is random. Moore and Spruill (1975) establish the corresponding statement for the unpenalized case; it is untested at $p/n = 0.25$;

(C4) p is fixed as $n \rightarrow \infty$.

Condition (C4) is what confines the results of this section to the fixed-dimension regime. It is not satisfied by design B, where $p/n = 0.25$, and this is exactly why the validity we claim there rests on prepivoting and on the simulation evidence of Section S4, not on Propositions 1 and 2. We are explicit about this in the discussion of the main text.

Remark 1. Conditions (C1)–(C3) are those of Moore and Spruill (1975) with one deletion and one addition. Moore and Spruill require the estimator to be $\sqrt{n}$ -consistent for β_0 and to satisfy an orthogonality condition on the estimating function. A penalized estimator satisfies neither when λ grows with n : it is $\sqrt{n}$ -consistent for a shrunken pseudo-parameter, not for β_0 . Proposition 1 is what replaces their conclusion once those two conditions are dropped.

S1.2 Proof of Proposition 1

Proposition 1. Under (C1)–(C4), $\hat{\beta} - \beta = M^{-1}\{X^\top(y - \pi) - K\beta\} + o_p(1)$ and $r \sim \mathcal{N}(\mu_K, \Omega_K)$ with

$$\begin{aligned}\Omega_K &= I_G - UM^{-1}U^\top - UM^{-1}KM^{-1}U^\top = I_G - UM^{-1}(F + 2K)M^{-1}U^\top, \\ \mu_K &= UM^{-1}K\beta + (\text{second-order remainder}).\end{aligned}$$

Proof. The penalized score is $s(\beta) = X^\top\{y - \pi(\beta)\} - K\beta$, and $\hat{\beta}$ solves $s(\hat{\beta}) = 0$. Expanding about the true β and using $-\partial s/\partial \beta = F + K = M$,

$$0 = s(\hat{\beta}) = X^\top(y - \pi) - K\beta - M(\hat{\beta} - \beta) + o_p(n^{1/2}),$$

which on rearranging gives the stated expansion, the remainder being $o_p(1)$ after scaling by $n^{-1/2}$ under (C2).

For the residuals, expand $\hat{\pi} = \pi(\hat{\beta})$ about $\pi = \pi(\beta)$: $\hat{\pi} = \pi + WX(\hat{\beta} - \beta) + o_p(n^{-1/2})$, so that

$$r = V^{-1/2}C(y - \hat{\pi}) = V^{-1/2}C(y - \pi) - U(\hat{\beta} - \beta) + o_p(1),$$

using $U = V^{-1/2}CWX$. Substituting the expansion for $\hat{\beta} - \beta$,

$$r = \underbrace{(V^{-1/2}C - UM^{-1}X^\top)}_{=:A}(y - \pi) + UM^{-1}K\beta + o_p(1). \quad (\text{S1})$$

Under H_0^{str} we have $\mathbb{E}(y - \pi) = 0$, so $\mathbb{E}(r) = UM^{-1}K\beta$, which is μ_K . We write the remainder in words rather than as an order symbol because $\|K\beta\| = O(\lambda)$, so $O(\|K\beta\|^2)$ is not a useful bound against a leading term of order $n^{-1/2}$. What we can say is measured rather than proved: at design A with $\lambda = 100$ the largest discrepancy between the empirical mean of r and μ_K is 0.1137 against $\|\mu_K\| = 3.871$, about 3%; and at $\lambda = 416$ in design B the second-order term overshoots the empirical mean by roughly a factor of two, which is why Section S2 row (b) fails.

For the variance, $\text{var}(y - \pi) = W$, so $\text{var}(r) = AWA^\top$. Expanding, and using the three identities $V^{-1/2}CW C^\top V^{-1/2} = I_G$, $V^{-1/2}CW X = U$ and $X^\top W X = F$,

$$AWA^\top = I_G - UM^{-1}U^\top - UM^{-1}U^\top + UM^{-1}FM^{-1}U^\top = I_G - 2UM^{-1}U^\top + UM^{-1}FM^{-1}U^\top.$$

Since $F = M - K$, $UM^{-1}FM^{-1}U^\top = UM^{-1}U^\top - UM^{-1}KM^{-1}U^\top$, and substituting gives $\Omega_K = I_G - UM^{-1}U^\top - UM^{-1}KM^{-1}U^\top$. For the second form, $UM^{-1}(F + 2K)M^{-1}U^\top = UM^{-1}(M + K)M^{-1}U^\top = UM^{-1}U^\top + UM^{-1}KM^{-1}U^\top$. Asymptotic normality follows from the Lindeberg–Feller theorem applied to the linear form in (S1) under (C1)–(C3). $\square$

Remark 2. Setting $K = 0$ recovers $I_G - UF^{-1}U^\top$, the classical Chernoff and Lehmann (1954); Moore and Spruill (1975) form. The ordering against that form is the opposite of what a first glance at Proposition 1 suggests, and it is worth deriving because the wrong sign here would misdirect the whole diagnosis. Since $M = F + K$ gives $F^{-1} - M^{-1} = F^{-1}KM^{-1} = M^{-1}KF^{-1}$, and $M^{-1}(F + 2K)M^{-1} = M^{-1}(M + K)M^{-1} = M^{-1} + M^{-1}KM^{-1}$,

$$F^{-1} - M^{-1}(F + 2K)M^{-1} = (F^{-1} - M^{-1}) - M^{-1}KM^{-1} = M^{-1}K(F^{-1} - M^{-1}) = M^{-1}KF^{-1}KM^{-1} \succeq 0,$$

the last matrix being a congruence of $F^{-1} \succ 0$. Hence $\Omega_K \succeq I_G - UF^{-1}U^\top$: the penalized covariance is *larger* than the maximum likelihood one, so a test referred to the classical covariance understates the residual variance and is anti-conservative on that account alone. Table S3 confirms it empirically — $\text{tr}(\Omega_K)$ exceeds $\text{tr}(\Omega_{\text{MLE}})$ in every row.

The covariance therefore pushes in the *same* direction as the non-centrality rather than against it. It is nevertheless the much smaller of the two effects — the trace gap is 0.23 in design A and 0.92 in design B, against a size distortion that reaches 1.000 — and Proposition 2 disposes of it entirely, since after $\hat{\mu}$ is subtracted the covariance is exactly the maximum likelihood one. The non-centrality μ_K is what has to be removed by hand.

S1.3 Proof of Proposition 2

Proposition 2. Let $\tilde{\beta} = \hat{\beta} + F^{-1}K\hat{\beta}$ and $\hat{\mu} = UM^{-1}K\tilde{\beta}$. Then, to first order, $\mathbb{E}(r - \hat{\mu}) = 0$ and $\text{var}(r - \hat{\mu}) = I_G - UF^{-1}U^\top$.

Proof. The key is that $\tilde{\beta}$ is exactly unbiased to first order. Since $I + F^{-1}K = F^{-1}(F + K) = F^{-1}M$,

$$\tilde{\beta} = (I + F^{-1}K)\hat{\beta} = F^{-1}M\hat{\beta} = F^{-1}M[\beta + M^{-1}\{X^\top(y - \pi) - K\beta\}] + o_p(1),$$

and expanding, $F^{-1}M\beta = F^{-1}(F + K)\beta = \beta + F^{-1}K\beta$, so the $F^{-1}K\beta$ terms cancel and

$$\tilde{\beta} = \beta + F^{-1}X^\top(y - \pi) + o_p(1). \quad (\text{S2})$$

That is, the one-step correction returns precisely the first-order expansion of the unpenalized maximum likelihood estimator, whether or not that estimator exists.

Hence $\hat{\mu} = UM^{-1}K\beta + UM^{-1}KF^{-1}X^\top(y - \pi) + o_p(1)$, and subtracting from (S1) the non-random terms cancel exactly:

$$\begin{aligned} r - \hat{\mu} &= (A - UM^{-1}KF^{-1}X^\top)(y - \pi) + o_p(1), \\ A - UM^{-1}KF^{-1}X^\top &= V^{-1/2}C - UM^{-1}(I + KF^{-1})X^\top. \end{aligned}$$

Now the cancellation that gives the proposition its content:

$$M^{-1}(I + KF^{-1}) = M^{-1}(F + K)F^{-1} = M^{-1}MF^{-1} = F^{-1}. \quad (\text{S3})$$

So $r - \hat{\mu} = A^*(y - \pi) + o_p(1)$ with $A^* = V^{-1/2}C - UF^{-1}X^\top$, whose mean is zero and whose variance is

$$A^*WA^{*\top} = I_G - UF^{-1}U^\top - UF^{-1}U^\top + UF^{-1}FF^{-1}U^\top = I_G - UF^{-1}U^\top. \quad \square$$

Remark 3. The estimation noise that the plug-in $\hat{\mu}$ injects is exactly the amount needed to convert the penalized covariance back into the maximum likelihood covariance. This is a cancellation, not an approximation: (S3) is an algebraic identity. It is the reason the practitioner does not have to learn a new reference distribution.

It also identifies the one quantity the procedure depends on badly, namely F^{-1} . When F is near-singular — light penalty with p/n large — the correction $F^{-1}K\hat{\beta}$ can overshoot. Section S4.3 measures this.

S1.4 Proof of Proposition 3

Proposition 3. *Let $(\eta^*, \hat{\eta}^*)$ be standard bivariate normal with correlation ρ . Then $\mathbb{E}\{\text{He}_k(\eta^*) \mid \hat{\eta}^*\} = \rho^k \text{He}_k(\hat{\eta}^*)$, and consequently a departure equal to the degree- k Hermite component of the true linear predictor contributes non-centrality attenuated by ρ^{2k} to a test that groups on the fitted predictor.*

Proof. Use the generating function $\sum_{k \geq 0} (t^k/k!) \text{He}_k(x) = \exp(tx - t^2/2)$. Given $\hat{\eta}^* = v$, we have $\eta^* \sim \mathcal{N}(\rho v, 1 - \rho^2)$, so

$$\begin{aligned} \mathbb{E}\{\exp(t\eta^* - t^2/2) \mid \hat{\eta}^* = v\} &= \exp(-t^2/2) \exp\{t\rho v + \tfrac{1}{2}t^2(1 - \rho^2)\} \\ &= \exp\{(t\rho)v - (t\rho)^2/2\} = \sum_{k \geq 0} \frac{(t\rho)^k}{k!} \text{He}_k(v). \end{aligned}$$

Comparing coefficients of t^k gives the conditional expectation. For the second claim, a grouped test on $\hat{\eta}^*$ is a function of the data only through group means, so the departure it responds to is $\mathbb{E}\{g(\eta^*) \mid \hat{\eta}^*\}$. Taking $g = \text{He}_k$ gives $\rho^k \text{He}_k(\hat{\eta}^*)$; the non-centrality of a quadratic-form statistic is proportional to the squared norm of that mean vector, hence to ρ^{2k} . $\square$

Remark 4. The visible effect size τ of the main text has $\text{sd}[\mathbb{E}\{g(x) \mid \hat{\eta}\}]$ in its numerator, which already contains the factor ρ^k . So τ absorbs the attenuation by construction, and no residual dimensional offset should be expected on a τ axis — as the index-aligned panel of the power figure in the main text confirms. Proposition 3 governs how much τ a given nominal γ buys, not the shape of power as a function of τ .

S2 Alternative corrections that fail

The prepivoted procedure of the main text has four neighbouring constructions that are simpler and that do not work. We report them because each failure is diagnostic, and because a reader’s first reaction to Proposition 2 is reasonably “why not just subtract $\hat{\mu}$ and use the χ^2 reference?”.

Three comments on Table S1.

(a) The analytic correction is enough in design A and not in design B. This is the boundary of the first-order theory, and it is the reason the paper does not stop at Proposition 2.

(b) It is natural to respond to (a) by taking the expansion one term further. This makes matters worse, not better, and the diagnostic shows why: at $\lambda = 416$ the first-order μ_1 tracks the empirical mean of the grouped residuals closely — the empirical mean ranges over $(-3.79, 3.65)$ and μ_1 over $(-3.41, 3.36)$ — whereas the second-order μ_2 ranges over $(-6.32, 6.53)$, roughly twice too large. The expansion is at its radius of convergence, so adding terms diverges rather than refines. This is the clearest single piece of evidence that the problem at $p/n = 0.25$ is not one that a better expansion solves.

(d) The generator matters as much as the statistic. Generating from the penalized fit builds the shrinkage into the null world, so the bootstrap statistics carry the same displacement as the observed one and the test loses its ability to reject. Generating from $\pi(\tilde{\beta})$ instead is what makes the null world the *structural* null of H_0^{str} of Section 2.2 of the main text, which is the hypothesis we mean to test, rather than the calibration null H_0^{cal} , which the penalized fit violates by construction.

Table S1: Rejection rate under a correctly specified model, nominal level 0.05, for the constructions that fail. Design B entries are given at the penalties stated. Every row was run on the same designs and groupings as the main results.

Construction	Rejection rate under H_0	Verdict
(a) Analytic corrected statistic referred to Ω_{MLE}	A: 0.036–0.064; 0.050/0.026/0.016 $\lambda = 20/50/416$	B: conservative in B at
(b) Second-order Taylor expansion of μ	A: 0.043–0.057; 0.240/0.107/0.103	B: overshoots by $\approx 2\times$
(c) Debiased bootstrap with the <i>plain</i> (uncorrected) statistic	0.000–0.060	conservative everywhere
(d) Prepivoting with the generator $\hat{\pi}_K$ from the <i>penalized</i> fit	0.008/0.025	conservative
(e) Prepivoting with generator $\pi(\tilde{\beta})$ and the corrected statistic in <i>both</i> worlds	see the corrected-size table of the main text	valid
(f) Wild Rademacher multiplier bootstrap of $r - \hat{\mu}$, refit-free	0.058/0.033	promising; one run only

Row (f) is a genuinely attractive alternative because it avoids refitting entirely, and therefore costs a fraction of the computation. We report it as promising rather than established: it rests on a single run at $B = 120$, which gives a Monte Carlo standard error of about 0.02, and we have not examined its power. It is the natural subject of follow-up work.

S3 Simulation designs

Design A ($n = 500$, $p = 5$). Covariates are independent standard normal. The linear predictor is $\eta_i = x_i^\top \beta_0$ with $\beta_0 = (0.5, -0.4, 0.3, -0.3, 0.2)^\top$ and no intercept, giving $\text{sd}(\eta) \approx 0.79$ and a balanced response. Penalties examined are $\lambda \in \{100, 137, 200\}$, equivalently $\lambda_g \in \{0.20, 0.274, 0.40\}$.

Design B ($n = 400$, $p = 100$, $p/n = 0.25$). Covariates are multivariate normal with an autoregressive correlation $\Sigma_{jk} = 0.7^{|j-k|}$. The coefficient vector cycles $(0.35, -0.30, 0.25, -0.20, 0.15)$ over the 100 coordinates and is scaled by 0.8887 so that $\|\beta_0\| = 2.31$. Penalties examined are $\lambda \in \{50, 416, 1000\}$, equivalently $\lambda_g \in \{0.125, 1.04, 2.5\}$.

In both designs the fitting is ridge logistic regression by iteratively reweighted least squares on the penalized normal equations, with the intercept unpenalized, to a tolerance of 10^{-9} and at most 60 iterations. The penalty scale identity $\lambda = n\lambda_g$ against `glmnet` was verified numerically to 1.6×10^{-9} .

Grouping is by equal-frequency deciles of the fitted probability throughout unless stated otherwise. The EDGE basis is the orthogonal cubic polynomial basis in the group-mean fitted probability, giving a three-dimensional quadratic form; the implementation used here was verified against `ebrahim.gof::edge.gof` to a difference of 0.00×10^0 on three datasets.

Under the null, data are generated from the design’s own logistic model, so every rejection reported as “size” is a false rejection of a model that is exactly correct.

Table S2: Rejection rate of the uncorrected tests under a *correctly specified* model, at nominal level 0.05, over the ridge penalty. This is the grid plotted in panels A and B of the penalty-path figure of the main text. Design A has $n = 500$ and $p = 5$; design B has $n = 400$ and $p = 100$ with an autoregressive covariate correlation of 0.7. Penalties are on the Friedman et al. (2010) scale $\lambda_g = \lambda/n$. Based on 500 replications, Monte Carlo standard error at most 0.022. The design B mean absolute error row is the quantity plotted on the horizontal axis of panel C.

Design A: λ_g	0	0.02	0.05	0.10	0.20	0.274
Hosmer–Lemeshow	0.068	0.068	0.118	0.362	0.916	0.988
EDGE basis	0.034	0.046	0.170	0.888	1.000	1.000
Design B: λ_g	0	0.01	0.05	0.124	0.513	1.5
Hosmer–Lemeshow	0.178	0.432	0.990	0.994	1.000	1.000
EDGE basis	0.284	0.994	1.000	1.000	1.000	1.000
MAE($\hat{\pi}, \pi_0$)	0.1859	0.1469	0.1330	0.1336	0.1606	0.1840

S4 Full grids and sensitivity

S4.1 Size of the uncorrected tests

Table S2 gives the headline grid. Two features of it are worth recording here. Over 500 replications of both designs, the one-standard-error cross-validation rule selected $\lambda_g > 0.05$ in *every* replication, so the columns at which the uncorrected test has already failed are the columns practice actually visits. And in design B the unpenalized column is itself invalid, at 0.178 and 0.284; this is the Sur and Candès (2019) and Candès and Sur (2020) phenomenon and it means that declining to penalize is not an escape.

S4.2 The effect of the correction on the residual geometry

Table S3: Effective degrees of freedom of the grouped residual vector, $G = 10$. “Removed” is $\text{tr}(UF^{-1}U^\top)$, the number of decile directions absorbed by estimation. “EDGE-3” is the effective dimension the three EDGE directions actually receive out of a nominal three. Averaged over 60 replications; these are geometric quantities and vary little between replications.

Design	λ	$\text{df}(\Omega_K)$	$\text{df}(\Omega_{\text{MLE}})$	Removed	EDGE-3	$\max \text{eig}(\Omega_{\text{MLE}})$
A	100	8.20	7.97	2.03	2.020	1.000
A	137	8.27	7.97	2.03	2.021	1.000
A	200	8.37	7.97	2.03	2.021	1.000
B	50	6.89	5.97	4.03	1.568	0.853
B	416	7.80	6.03	3.97	1.546	0.845
B	1000	8.19	6.04	3.96	1.539	0.844

Table S3 quantifies what dimension costs a grouped test, independently of any particular alternative.

(a) Estimation absorbs 2.03 of the ten decile directions in design A and 4.03 in design B — almost exactly twice as many. A test with ten groups is effectively a six-dimensional test at $p/n = 0.25$.

(b) In design A the largest eigenvalue of Ω_{MLE} is exactly 1.000: some direction of the grouped residual vector is left entirely untouched by estimation. In design B the largest eigenvalue never exceeds 0.853. No direction survives intact, which is a compact way of saying what p/n costs.

(c) The three EDGE directions receive only about 1.55 effective dimensions in design B against 2.02 in design A.

S4.3 Behaviour of the debiasing at light penalty

Proposition 2 depends on F^{-1} , which is poorly conditioned when the penalty is light and p/n is large. Table S4 measures the consequence.

Table S4: Norm of the debiased estimate relative to the truth, averaged over 60 replications. The debiasing recovers the scale of β_0 almost exactly except at the lightest penalty in design B, where it overshoots by 26%.

Design	λ	$\ \beta_0\ $	$\ \hat{\beta}\ $	$\ \tilde{\beta}\ /\ \beta_0\ $
A	100	0.79	0.42	0.99
A	137	0.79	0.36	0.98
A	200	0.79	0.29	0.96
B	50	2.31	0.84	1.26
B	416	2.31	0.27	1.05
B	1000	2.31	0.17	1.02

The overshoot is confined to design B at $\lambda = 50$ and is within 5% of the truth in all five other cells, which is what the conditioning argument predicts: heavier penalties make $M = F + K$ better conditioned, so the correction is stable again.

We tested whether the overshoot is consequential by intervening on it directly, which is a stronger design than correlating it with anything. Holding the statistic, the grouping and the observed data fixed, we replaced the generator $\pi(\tilde{\beta})$ by $\pi\{\hat{\beta} + s(\tilde{\beta} - \hat{\beta})\}$ with $s = 0.71$, chosen so that the generator has the same norm as β_0 ; the achieved ratio moved from 1.26 to 1.00. Over 400 replications at $\lambda = 50$ the rejection rate on the EDGE basis was 0.0500 before the intervention and 0.0550 after it, a difference well inside the Monte Carlo standard error of 0.011, and the ratio of bootstrap to observed statistic variance moved from 0.72 to 0.74. Removing the overshoot does not remove the conservatism.

Two cautions about how far this intervention settles the question, because it does not settle it as cleanly as the arithmetic above suggests. First, the pre-intervention arm returned 0.0500, whereas the pooled size of that same cell over 4000 replications is 0.0370; the 400-replication block used here did not itself display the conservatism the intervention was meant to remove. Second, the quoted 0.011 is an unpaired Monte Carlo standard error, whereas the design is paired — the statistic, the grouping and the data are held fixed — so the experiment’s true resolution is finer than 0.011 and we have not measured it. What can be said without either caveat is that the bootstrap reference is *less* dispersed than the observed statistic, $\text{var}(S^*)/\text{var}(S) = 0.72$, which pushes towards over-rejection and so cannot account for under-rejection either.

We therefore record the overshoot as an unexplained correlate rather than a refuted hypothesis. It is large, it is confined to exactly the anomalous cell, and it has a clean mechanism in the conditioning of F ; what we can show is that removing it does not remove the conservatism. The overshoot is a real caution about the behaviour of $\tilde{\beta}$ at light penalty when p/n is large, and it is worth knowing for anyone who wants to use $\tilde{\beta}$ for its own sake, but it does not affect the size of the test and we do not offer it as an account of Section S4.6’s residual conservatism, which remains unexplained.

S4.4 Size-adjusted power

Because the EDGE basis under-rejects in design B at light penalty (Section S4.6), a comparison at the nominal level is not quite a comparison of like with like. We therefore repeated every power comparison at a size-adjusted level: the $\gamma = 0$ cell of each panel supplies the null distribution of the p -value, and the alternative cells are thresholded at its fifth percentile. The thresholds were 0.0533 in design A on both bases, and 0.0400 (decile) and 0.0533 (EDGE) in design B.

Table S5: Nominal and size-adjusted power, 2000 replications per cell. The adjustment moves no entry by more than 0.02 and reverses no comparison, so the conclusions of the power table of the main text do not depend on it.

Design	Departure	γ	Nominal		Size-adjusted	
			decile	EDGE	decile	EDGE
A	—	0.00	0.0445	0.0460	0.0520	0.0530
A	quadratic	0.25	0.0760	0.1150	0.0820	0.1305
A	quadratic	0.50	0.1475	0.2615	0.1660	0.2820
A	quadratic	0.75	0.2015	0.3755	0.2175	0.3940
A	quadratic	1.00	0.2700	0.4150	0.2885	0.4300
A	interaction	0.25	0.0595	0.0750	0.0680	0.0830
A	interaction	0.50	0.1055	0.1885	0.1185	0.2045
A	interaction	0.75	0.1940	0.3555	0.2140	0.3745
A	interaction	1.00	0.2835	0.4615	0.3080	0.4855
B	—	0.00	0.0640	0.0495	0.0560	0.0580
B	quadratic	0.50	0.0595	0.0485	0.0535	0.0570
B	quadratic	1.00	0.0635	0.0600	0.0525	0.0670
B	quadratic	1.50	0.0740	0.0630	0.0655	0.0675

S4.5 Computational cost of the procedure and of prepivoting

The obvious objection to a procedure that refits inside a bootstrap loop is its price, so we state it. On the machine and worker count given in Table S8, one prepivoted test at $N_B = 149$ takes about 0.26 seconds in design A and about 3.3 seconds in design B, the difference being the cost of 149 ridge refits at $p = 100$. Two independent timings agree to within 2%. For a single analysis — which is what a practitioner runs — the cost is seconds, and the simulation studies in this paper are expensive only because they repeat that analysis thousands of times.

Prepivoting itself is not the expensive part, and it is not a statistical tax either. In design A at $\lambda = 100$, referring the corrected statistic to the analytic Ω_{MLE} reference gives rejection rates 0.023, 0.147 and 0.227 at $\gamma = 0, 0.5$ and 1.0 , while prepivoting the same statistic gives 0.040, 0.163 and 0.250. Prepivoting is closer to nominal under the null *and* higher under both alternatives. This is a smaller run than the size study — 300 replications, so a Monte Carlo standard error of about 0.029 — and we would not read the individual gaps as established; the point is only that resampling does not buy validity at the price of power here.

S4.6 Re-estimation of the contested cell

The EDGE basis in design B at $\lambda = 50$ is the one cell of the corrected-size table of the main text that we estimated more than once, and we set out the sequence because it bears on how such tables should be read.

Our first estimate, from a single block of 1000 replications, was 0.0270 with a standard error of 0.0051: four and a half standard errors below nominal, and outside the range $[0.03, 0.08]$ that we had fixed in advance as acceptable. Before writing about it we re-estimated the cell on two further blocks of 1500, disjoint from the first and from each other. Table S6 gives all three.

Three conclusions. The conservatism on the EDGE basis is *real*: pooled over 4000 replications the size is 0.0370, with an exact 95% interval of $[0.0314, 0.0433]$ and $z = -4.36$ against nominal, so it is not the kind of blemish that disappears with more computation. Its *magnitude* was nonetheless overstated by the first block by about a third. And the pooled value lies inside $[0.03, 0.08]$, so the qualitative verdict on the cell changes even though the direction does not.

Table S6: The contested cell, design B at $\lambda = 50$, estimated on three independent replication blocks. The decile basis was never in doubt. On the EDGE basis the first block is the lowest of the three; the three are homogeneous and the pooled estimate lies inside the acceptable range.

Basis	Block 1 (1000)	Block 2 (1500)	Block 3 (1500)	Homogeneity	Pooled (4000)
Decile	0.0630	0.0587	0.0580	$p = 0.86$	0.0595 (0.0037)
EDGE	0.0270	0.0440	0.0367	$p = 0.09$	0.0370 (0.0030)

We draw the obvious methodological moral. A rejection rate near 0.03 estimated on 1000 replications has a standard error of 0.005, so blocks will disagree by more than one might expect, and the residual heterogeneity here ($p = 0.09$) suggests 1000 is thin for a tail probability this small. Any cell that falls outside a pre-specified range should be re-estimated on an independent block before conclusions are drawn from it.

S4.7 Sensitivity to the number of groups and to the number of bootstrap replicates

On the glaucoma data the corrected p -value varies from 0.026 at $G = 10$ to 0.172 at $G = 20$ on the decile basis. This sensitivity is inherited from the Hosmer–Lemeshow family and the correction does not remove it; the full $\lambda \times G$ grid is the full $\lambda \times G$ glaucoma table of the main text, and we report all twelve cells rather than a selected one. On the same data, repeating the cross-validated cell at $G = 10$ with $N_B = 999$ under different seeds gave corrected p -values in the range 0.015–0.026 on the decile basis and 0.034–0.049 on the EDGE basis, so the reported figures are stable to the bootstrap size.

S5 The glaucoma analysis in detail

The GlaucomaM data are distributed in the R package TH.data. There are $n = 196$ eyes, 98 glaucomatous and 98 normal, matched by age and sex, with $p = 62$ morphometric variables derived from confocal laser scanning tomography of the optic nerve head. All variables were centred and scaled before fitting.

Design facts: $\text{cond}(X^\top X) = 4.97 \times 10^7$, largest absolute pairwise correlation 0.9961, and $p/n = 0.316$. The unpenalized logistic fit does not converge; the maximum likelihood estimate does not exist. Ten-fold cross-validation gives $\hat{\lambda}_{\min} = 19.9$ and $\hat{\lambda}_{\text{lse}} = 295.3$ on the theory scale, the latter corresponding to $\lambda_g = 1.51$. At $\hat{\lambda}_{\text{lse}}$ the fit has area under the curve 0.905, $\text{sd}(\hat{\pi}) = 0.224$ and fitted probabilities spanning $[0.016, 0.911]$; at $\hat{\lambda}_{\min}$ the area under the curve is 0.942. The apparent calibration slope at $\hat{\lambda}_{\text{lse}}$ is 2.434.

S5.1 A second dataset

The phenomenon is not confined to one dataset. GlaucomaMVF (package ipred) is a second confocal-tomography series, $n = 170$ eyes and $p = 63$ morphometric variables, so $p/n = 0.371$; here too the maximum likelihood estimate does not exist. Cross-validation gives $\hat{\lambda}_{\min} = 10.2$ and $\hat{\lambda}_{\text{lse}} = 54.6$ on the theory scale. Table S7 reports the same grid we ran for GlaucomaM.

Three things should be said about Table S7.

(a) The reversal is cleaner here than on GlaucomaM. The corrected test returns p between 0.224 and 0.998 on the decile basis and between 0.298 and 0.874 on the EDGE basis, rejecting in none of the twelve cells, while the uncorrected EDGE test rejects in all twelve. Repeating the cross-validated cell at $G = 10$ with $N_B = 999$ under five seeds gives corrected p -values in $[0.883, 0.916]$ on the decile basis and $[0.419, 0.465]$ on the EDGE basis, so the conclusion is not an artefact of the bootstrap size.

(b) The uncorrected *decile* test does not reject in four of the twelve cells, shown in bold, three of them at $G = 20$. We therefore do not claim that the uncorrected test rejects everywhere on this dataset;

Table S7: The uncorrected and corrected tests on GlaucomaMVF, four penalties by three group counts, 499 bootstrap replicates. The reversal is complete: the uncorrected EDGE test rejects in all twelve configurations, and the corrected test rejects in none of them on either basis.

λ	Uncorrected (dec / EDGE)			Corrected (dec / EDGE)		
	$G = 5$	$G = 10$	$G = 20$	$G = 5$	$G = 10$	$G = 20$
$\hat{\lambda}_{\min}$	0.077 / 0.041	0.023 / 0.003	0.218 / 0.005	0.458 / 0.318	0.556 / 0.350	0.476 / 0.298
$\frac{1}{2}\hat{\lambda}_{1se}$	0.007 / 0.013	0.029 / 0.003	0.335 / 0.003	0.224 / 0.826	0.706 / 0.552	0.808 / 0.636
$\hat{\lambda}_{1se}$	0.000 / 0.000	0.016 / 0.001	0.327 / 0.001	0.688 / 0.874	0.876 / 0.470	0.998 / 0.428
$2\hat{\lambda}_{1se}$	0.000 / 0.000	0.001 / 0.000	0.027 / 0.000	0.430 / 0.852	0.618 / 0.564	0.840 / 0.426

the claim holds on the EDGE basis, and on the decile basis at the finer groupings.

(c) We report this dataset in the supplement rather than the main text because it adds corroboration rather than a new argument. A reader who doubts that GlaucomaM is representative should read it as the reply.

We also examined Sonar (package `mlbench`; $n = 208$, $p = 60$, $p/n = 0.288$, maximum likelihood separates) and observed the same qualitative reversal, but only at a single penalty and group count. We do not report its p -values, because a single configuration cannot be distinguished from a favourable choice of one.

We also examined `wpbc` and rejected it as an example. At the cross-validated penalty the largest fitted coefficient is 0.00099 and the fitted probabilities span $[0.234, 0.242]$: cross-validation had selected an essentially intercept-only model, so its apparent clean reversal from $p = 0.0008$ to $p = 0.80$ is an artefact of having no model to test rather than a demonstration of the method. We report this because it is the kind of example that would flatter the method if reported uncritically.

S6 Reproducibility

All computation is in R. Penalized fits use the iteratively reweighted least squares routine described in Section S3; `glmnet` (Friedman et al., 2010) is used for cross-validation and for the penalty-scale check. Parallel runs use PSOCK clusters with `parallel::clusterSetRNGStream` for reproducible streams; the stream seed is given with each result below.

Replication-level p -values are archived for the power study (`T25_power_pvalues.csv`) and for the head-to-head comparison with the high-dimensional alternatives (`T24_rivals_pvalues.csv`), so that every cell of the power tables and every size-adjusted threshold can be re-derived and its Monte Carlo standard error recomputed independently. The remaining experiments archive per-replication rejection indicators and the saved fitted objects (`*_results.rds`), which is enough to reproduce each reported rate but not to re-threshold it at a different level. The complete archive, including these files, the figure scripts and `_ek_theme.R`, is available at <https://github.com/ebrahimkhaled/gof-penalized-paper> and deposited at <https://doi.org/10.5281/zenodo.21903202>.

References

- Candès, E.J. and Sur, P. (2020). The phase transition for the existence of the maximum likelihood estimate in high-dimensional logistic regression. *The Annals of Statistics*, **48**(1), 27–42.
- Chernoff, H. and Lehmann, E.L. (1954). The use of maximum likelihood estimates in χ^2 tests for goodness of fit. *The Annals of Mathematical Statistics*, **25**(3), 579–586.
- Friedman, J., Hastie, T. and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, **33**(1), 1–22.

Table S8: Scripts, seeds and cost for every reported experiment.

Result	Script	Seed / replication	Cost
Proposition 1 numerical check	corrected_test.R	fixed design	seconds
Uncorrected size grid (Table 1)	lambda_sweep.R, hd_sweep.R	$B = 500$	minutes
Corrected size (Table 2)	T23_rerun.R	stream 20260807, seeds 1001–2000	11.2 min, 22 workers
Contested cell (Table S6)	verify_E13.R	seeds 1–1500 and 100001–101500	15 min, 22 workers
Power (Table 3, Table S5)	T25_power_highB.R	stream 20260810, $B = 2000$	37 min, 22 workers
Visible effect size (Figure 1)	T22_tau.R, plot_tau.R, R/fig1_power_ek.R	known-null MC, $B = 2000$ null, $B_1 = 1000$	minutes
Oracle grouping (Table 5)	oracle_knownnull.R	$B_0 = 2000$, $B_1 = 1000$, both arms exact	minutes
Residual geometry (Table S3)	omega_spectrum.R	stream 20260808, 60 reps	26 s, 22 workers
Overshoot intervention (§S4.3)	overshoot_test.R	stream 20260809, 400 reps	244 s, 22 workers
Failed repairs (Table S1)	round2.R, cand_C.R	$B = 120$ –300	minutes
Glaucoma analysis	glaucoma_deep.R, R/fig3_calib_ek.R	set.seed(1), $N_B = 499$	minutes
Second dataset (§S5.1)	T26_app.R, extract_T26b.R	set.seed(1), $N_B = 499$	minutes
Cost, and the price of pre pivoting	prepivot_cost.R	$B = 300$, design A $\lambda = 100$	minutes
Conditioning on the application	glaucoma_conditioning.R	set.seed(1)	seconds
Generator diagnostic	check_generator.R	set.seed(1)	seconds
Covariance ordering	check_omega_sign.R	500 random (F, K) draws	seconds

- Moore, D.S. and Spruill, M.C. (1975). Unified large-sample theory of general chi-squared statistics for tests of fit. *The Annals of Statistics*, **3**(3), 599–616.
- Sur, P. and Candès, E.J. (2019). A modern maximum-likelihood theory for high-dimensional logistic regression. *Proceedings of the National Academy of Sciences*, **116**(29), 14516–14525.